

Modeling of Thin Films and Cover Plates Bonded to Graded Substrates

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Abstract. In this study the contact problems for thin films and cover plates will be considered. In these problems the loading consists of any one or combination of stresses caused by uniform temperature changes and temperature excursions, far field mechanical loading, and residual stresses resulting from film processing or welding. The primary interest in this study is in examining stress concentrations or singularities near the film ends for the purpose of addressing the question of crack initiation and propagation in the substrate or along the interface. The underlying contact mechanics problem is formulated and solved analytically by reducing it to an integral equation.

Keywords: contact mechanics, thin films, friction, functionally graded materials.

INTRODUCTION

Generally the mechanics of contact problems involving the functionally graded materials may be studied in two broad categories. The first is the standard load transfer problems with friction in which the loading component may be rigid or deformable and the loaded component is usually deformable. As examples for such problems one may mention bearings, gears, cams, cylinder linings/piston rings, and abradable seals in stationary gas turbines. In these applications the concept of material property grading is used by coating the mechanical components with appropriate FGM layers, usually to improve wear resistance. The second category of contact problems consists of thin films and cover plates bonded to FGM substrates. The main applications of cover plates are in aerospace and civil engineering structures in which they are used as reinforcements or stiffeners, whereas thin films are used mostly in microelectronics.

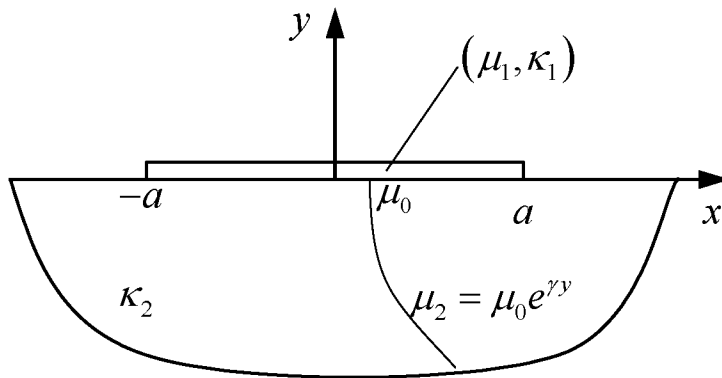


FIGURE 1. Geometry of the problem

In this study only the contact problems for thin films and cover plates will be considered. In these problems the loading consists of any one or combination of stresses caused by uniform temperature changes and temperature excursions, far field mechanical loading, and residual stresses resulting from film processing or welding. A characteristic feature of this second category of contact problems is that generally the film or cover plate thickness is very small compared to other dimensions in the system. Consequently, in formulating the mechanics problem it will be assumed that the substrate may be approximated by a semi-infinite graded elastic continuum and the film by an elastic “membrane”. Depending on the nature of the bonding process the interfacial zone may be modeled by either an ideal interface or a homogeneous shear layer.

The primary interest in this study is in examining stress concentrations or singularities near the film ends for the purpose of addressing the question of crack initiation and propagation in the substrate or along the interface. Among the many studies on singularities for edge debond problems [1–7], Erdogan and Gupta [1] is one of the earliest and most relevant to thin films, where singular shear stresses between an elastic stiffener and a half plane are determined by solving a compatibility equation in the form of a singular integral equation. Later, Shield and Kim [8] extended this analysis by imparting bending stiffness to the film to incorporate normal tractions in addition to interfacial shear. Alcala et al. [9] calculated interfacial shear stress between a rectangular strip of thin film and its substrate in closed form.

For the two interface models considered this is done by varying the film/substrate contact angle. The underlying contact mechanics problem is formulated and solved analytically by reducing it to an integral equation. For the ideal interface model it is shown that the integral equation has a simple Cauchy kernel. Hence, the contact stress has a standard square-root singularity. In the case of film/substrate bonding through an adhesive layer, the integral equation has only a logarithmic singularity and consequently, all stress components are bounded.

FORMULATION OF THE PLANE ELASTICITY PROBLEM

Consider the plane elasticity problem show in Figure 1. Medium 1 is a homogeneous thin film with thickness h , medium 2 is the graded half plane. The shear moduli of the thin film and substrate are given by μ_1 and $\mu_2(y)$ respectively.

μ_1 is constant and $\mu_2(y)$ is approximated by

$$\mu_2(y) = \mu_0 e^{\gamma y}, \quad y < 0, \quad (1)$$

where γ is a constant characterizing the material inhomogeneity, μ_0 is the value of $\mu_2(y)$ at the surface.

In the graded medium $-\infty < y < 0$ the spatial variation of the Poisson's ratio is assumed to be negligible. Thus, we have $\nu_2 = \nu_1(y) = \nu = \text{constant}$.

For the plane contact problem under consideration the Hooke's law in the region $-h < y < 0$ can be written as

$$\sigma_{2xx}(x, y) = \frac{\mu_2(y)}{\kappa_2 - 1} \left[(\kappa_2 + 1) \frac{\partial u_2}{\partial x} + (3 - \kappa_2) \frac{\partial v_2}{\partial y} \right], \quad (2a)$$

$$\sigma_{2yy}(x, y) = \frac{\mu_2(y)}{\kappa_2 - 1} \left[(3 - \kappa_2) \frac{\partial u_2}{\partial x} + (\kappa_2 + 1) \frac{\partial v_2}{\partial y} \right], \quad (2b)$$

$$\tau_{2xy}(x, y) = \mu_2(y) \left[\frac{\partial u_2}{\partial y} + \frac{\partial v_2}{\partial x} \right], \quad (2c)$$

where $\kappa_2 = 3 - 4\nu$ for plane strain and $\kappa_2 = (3 - \nu)/(1 + \nu)$ for the generalized plane stress conditions. Substituting (2) into the equilibrium equations we obtain

$$(\kappa_2 + 1) \frac{\partial^2 v_2}{\partial y^2} + (\kappa_2 - 1) \frac{\partial^2 v_2}{\partial x^2} + 2 \frac{\partial^2 u_2}{\partial x \partial y} + \gamma (3 - \kappa_2) \frac{\partial u_2}{\partial x} + \gamma (\kappa_2 + 1) \frac{\partial v_2}{\partial y} = 0, \quad y < 0, \quad (3a)$$

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By using the Fourier transforms and after some lengthy analysis we can write the x derivative of the displacement components, $u_2(x, y)$ and $v_2(x, y)$ as

$$-\omega \sigma_{2xy}(x, 0) + \frac{1}{\pi} \int_{-\infty}^{\infty} \left[\frac{1}{t-x} - k_{11}(t, x) \right] \sigma_{2xy}(t, 0) dt - \frac{1}{\pi} \int_{-\infty}^{\infty} \sigma_{2xy}(t, 0) k_{12}(t, x) dt = f(x), \quad -a < x < b, \quad (4a)$$

$$\omega \sigma_{2yy}(x, 0) + \frac{1}{\pi} \int_{-\infty}^{\infty} \left[\frac{1}{t-x} - k_{21}(t, x) \right] \sigma_{2yy}(t, 0) dt - \frac{1}{\pi} \int_{-\infty}^{\infty} \sigma_{2yy}(t) k_{22}(t, x) dt = g(x), \quad -a < x < b, \quad (4b)$$

where

$$f(x) = \lambda \frac{\partial}{\partial x} v_2(x, 0), \quad g(x) = \lambda \frac{\partial}{\partial x} u_2(x, 0), \quad \lambda = \frac{4\mu_0}{\kappa_2 + 1}, \quad \omega = \frac{\kappa_2 - 1}{\kappa_2 + 1}, \quad (5a, d)$$

and k_{ij} are known bounded functions(see Guler [10]).

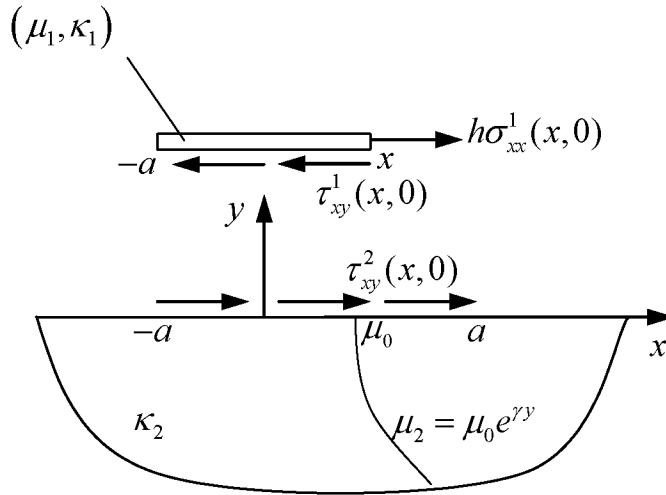


FIGURE 2. Force balance at the interface between the thin film and the FGM.

Under the following conditions,

$$\sigma_{2,xy}(x) = \begin{cases} 0 & |x| > a, \\ f(x) & |x| < a, \end{cases} \quad (6a,b)$$

$$\sigma_{2,yy}(x) = 0,$$

we can write the strain in the FGM from equation (4b) as

$$\frac{\partial}{\partial x} u_2(x, 0) = \frac{\kappa_2 + 1}{4\mu_2} \frac{1}{\pi} \int_{-\infty}^{\infty} \left[\frac{1}{t-x} + k_{21}(t, x) \right] \sigma_{2,xy}(t, 0) dt. \quad (7)$$

Equilibrium of the cover plate requires (see Fig. 2.)

$$\int_{-a}^x \sigma_{1,xy}(t, 0) dt = \int_{-a}^x f(t) dt = \sigma_{1,xx}(x, 0)h. \quad (8)$$

If we consider a perfect bonding between the thin film and the FGM, compatibility condition at $y = 0$ becomes

$$\varepsilon_{1,xx}(x, 0) = \varepsilon_{2,xx}(x, 0). \quad (9)$$

For the plane strain conditions

$$\sigma_{1,xx} = \frac{8\mu_1}{1 + \kappa_1} \frac{\partial}{\partial x} u_1(x, 0), \quad (10)$$

and the equilibrium equation (8) becomes

$$\int_{-a}^x f(t) dt = \frac{8\mu_1}{1 + \kappa_1} \frac{\partial}{\partial x} u_1(x, 0)h, \quad (11a)$$

$$\int_{-a}^a f(t) dt = 0. \quad (11b)$$

Therefore strain in the thin film can be written from equation (11a) as

$$\frac{\partial}{\partial x} u_1(x, 0) = \frac{1 + \kappa_1}{8\mu_1} \frac{1}{h} \int_{-a}^x f(t) dt. \quad (12)$$

Superposing the strains caused by the external loads other than $f(x)$, equation (12) becomes

$$\frac{\partial}{\partial x} u_2(x, 0) = \frac{\kappa_2 + 1}{4\mu_2} \frac{1}{\pi} \int_{-\infty}^{\infty} \left[\frac{1}{t-x} + k_{21}(t, x) \right] f(t) dt + g(x), \quad (13)$$

where $g(x)$ is the resulting strain from the external loads other than $f(x)$.

Using (12) and (13) equation (9) becomes

$$\frac{\partial}{\partial x} u_1(x, 0) = \frac{\partial}{\partial x} u_2(x, 0), \quad (14a)$$

$$\frac{1}{\pi} \int_{-a}^a \left[\frac{1}{t-x} + k_{21}(t, x) \right] f(t) dt - \lambda \int_{-a}^x f(t) dt = -\frac{4\mu_2}{\kappa_2 + 1} g(x), \quad (14b)$$

where

$$\lambda = \frac{\pi}{2h} \frac{1 + \kappa_1}{1 + \kappa_2} \frac{\mu_2}{\mu_1}, \quad g(x) = \varepsilon_0 + \alpha T. \quad (15a, b)$$

Or in short form

$$\frac{1}{\pi} \int_{-a}^a \left[\frac{1}{t-x} + k_{21}(t, x) - \lambda k_F(t, x) \right] f(t) dt = -\frac{4\mu_2}{\kappa_2 + 1} g(x), \quad (16)$$

where

$$k_F(x, t) = H(t+a) - H(t-x) = \begin{cases} 1 & -a < t < x, \\ 0 & x < t < a. \end{cases} \quad (17)$$

SOLUTION OF THE INTEGRAL EQUATION

By defining the following normalized quantities

$$x = ar, \quad t = as, \quad -a < (x, t) < a, \quad -1 < (r, s) < 1, \quad (18a, c)$$

$$f(x) = cg(r)p(r), \quad c = -\frac{4\mu_2}{\kappa_2 + 1}, \quad (19d, e)$$

the integral equation (16) and equation (11b) may be expressed in the following form

$$\frac{1}{\pi} \int_{-1}^1 \left[\frac{1}{s-r} + k_{21}(s, r) - \lambda k_F(s, r) \right] p(s) ds = 1, \quad -1 < r < 1, \quad (20)$$

$$\frac{1}{\pi} \int_{-1}^1 p(s) ds = 0, \quad (21)$$

where the kernel $k_{21}(s, r)$ is given by

$$k_{21}(s, r) = ak_{21}(t, x). \quad (22)$$

Assuming a solution of the form

$$p(s) = \phi(s) \frac{1}{\sqrt{1-s^2}}, \quad (23)$$

where

$$\phi(s) = \sum_0^{\infty} a_n T_n(s). \quad (24)$$

Using the orthogonality conditions of the Chebyshev polynomials

$$\frac{1}{\pi} \int_{-1}^1 \frac{T_n(s) T_m(s)}{\sqrt{1-s^2}} ds = \begin{cases} 1 & m = n = 0, \\ 1/2 & m = n \geq 1, \\ 0 & m \neq n. \end{cases} \quad (25)$$

and the properties of the Chebyshev polynomials

$$\frac{1}{\pi} \int_{-1}^1 \frac{T_n(s)}{(s-r)\sqrt{1-s^2}} ds = \begin{cases} 0, & n = 0, \\ U_{n-1}(r), & n > 0, \end{cases} \quad (26)$$

it may be seen that $a_0 = 0$, and the integral equation (20) may be reduced to

$$\sum_1^{\infty} a_n [U_{n-1}(r) + R_n(r)] = 1, \quad -1 < r < 1 \quad (27)$$

where

$$R_n(r) = \frac{1}{\pi} \int_{-1}^1 \frac{T_n(s)}{\sqrt{1-s^2}} \left[k_{21}(s, r) - \frac{\lambda}{\pi} k_F(s, r) \right] ds. \quad (28)$$

Therefore the shear stress at the interface $y = 0$

$$\sigma_{2xy}(x, 0) = f(x) = -\frac{4\mu_2}{\kappa_2 + 1} g(r) \sum_0^{\infty} a_n T_n(s) \frac{1}{\sqrt{1-s^2}}, \quad (29)$$

where

$$g(x) = \varepsilon_0 + \alpha T = \frac{\sigma_0}{E_2} + \alpha T. \quad (30)$$

SUMMARY

This study presents the formulation of the problem of a thin film bonded to a graded substrate. Using Fourier transforms all the boundary conditions are satisfied analytically and the problem is reduced to a singular integral equation. A numerical solution procedure that employs Chebyshev polynomials is developed and the solution for the interface shear stress is derived.

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